16 Make t the subject of the formula $k = \dfrac{2(t+3)}{t-3}$

$k(t - 3) = 2(t + 3)$

1 mark
Multiply by(t−3)

$kt - 3k = 2t + 6$

$kt - 2t = 3k + 6$

1 mark
Gather t terms

$t(k - 2) = 3k + 6$

1 mark
Factorise t

$t = \dfrac{3k + 6}{k - 2}$

1 mark
(3k+6)/(k−2)

$k \neq 2$; *original denominator requires* $t \neq 3$.

(Total for Question 16 is 4 marks)